

Venkata Mani Sai Praneeth Chidella

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SUMMARY

AI Engineer with 4+ years of experience building production LLM systems, fraud detection pipelines, and RAG-based applications. Contributed to search, personalization, and MLOps infrastructure at Walmart; independently engineered a full-stack AI insurance claims platform (Smart Claim AI) combining GPT-4 Vision and Claude API with RAG; and published two peer-reviewed papers in NLP. Hands-on across the full ML lifecycle - data engineering, model training, evaluation (RAGAS/G-Eval), and deployment using GPT-4o, Claude 3.5 Sonnet, and local SLMs across fraud detection, claims processing, and NLP classification tasks.

CORE SKILLS

Languages: Python (Advanced) - Java - JavaScript - SQL - AI & LLM Orchestration: LangGraph - LangChain - Multi-Agent Systems - GPT-4o/Vision - Claude 3.5 Sonnet - OI- - lama - Prompt Engineering (CoT - ReAct) - Structured Output Validation (Pydantic) - LangSmith - Machine Learning: Scikit-learn - XGBoost - LightGBM - Random Forest - SVM - Feature Engineering - SMOTE/ADASYN - Explainable AI (SHAP - LIME) - Model Drift Monitoring - Deep Learning & NLP: PyTorch - CNN - LSTM - Transformers - BERT

EXPERIENCE

Walmart, USA Aug 2024 - Present

AI Engineer

Contributed to LLM-based search and personalization pipelines for product discovery; team achieved 35% reduction in average search latency and 18% improvement in click-through rates.

Built and maintained ML models for inventory management using historical sales and real-time demand signals, helping

reduce stockout incidents by 22% and excess holding costs by 15%.

Supported MLOps workflows on Vertex AI and Kubernetes, configuring automated retraining triggers and deployment

pipelines to reduce manual intervention and contributing to 99.8% uptime for production AI services.

Implemented model performance monitoring and alerting using custom evaluation metrics, enabling early detection of

data drift and degradation before impacting production traffic.

Developed embedding-based components for collaborative filtering recommendation systems; contributed to a 12% lift

in conversion rate and 9% increase in session engagement.

Built a computer vision module using YOLO and OpenCV for retail shelf monitoring - detecting out-of-stock items

and compliance issues - deployed on edge AI devices, reducing manual shelf audit time by 45%.

Partnered with product and data engineering teams to define evaluation metrics and track model performance post-deployment.

Maxgen Technologies, India Dec 2021 - Dec 2023

AI Engineer

Engineered a real-time fraud detection engine using an ensemble of Random Forest and XGBoost, processing 10,000+

daily transactions at 94% precision.

Built a high-dimensional feature pipeline using SQL and Pandas to extract 50+ behavioral indicators, improving detection of sophisticated fraud patterns by 20%.

Resolved extreme class imbalance using cost-sensitive learning and SMOTE/ADASYN, achieving 0.89 AUC-ROC on

production transaction data.

Integrated SHAP and LIME for explainable AI outputs, generating human-readable reasoning codes for flagged

trans-

actions to streamline manual audit workflows.

Deployed fraud models as Dockerized FastAPI microservices, maintaining sub-100ms response times under peak load.

Built an MLflow + CI/CD retraining loop with drift monitoring and A/B testing, reducing false negatives by 60% over 12 months.

Built an OCR pipeline using Tesseract, Hugging Face Transformers, and Azure Cognitive Services to extract and classify data from invoices and contracts, achieving 91% extraction accuracy and cutting manual data entry by 55%.

Smart Claim AI, USA Jan 2024 - Jul 2024

AI Engineer (Startup - Founding Contributor)

Independently built the full-stack foundation of an AI-powered insurance claims platform using React, Node.js/Express,

and MongoDB Atlas, deployed on Vercel and Render with OAuth authentication (Google, Facebook, GitHub).

Integrated GPT-4 Vision to analyze uploaded damage images - identifying damage type, affected areas, and severity - and connected it with the Claude API to generate structured, adjuster-ready claim reports.

Designed and implemented a RAG pipeline ingesting jurisdiction-specific insurance policy documents with semantic chunking and vector retrieval, grounding all AI outputs and eliminating hallucinated claim decisions.

Enforced output safety by restricting AI responses strictly to verified regulatory and policy content, ensuring no incorrect or sensitive information reached end users in a high-stakes legal context.

Orchestrated two AI APIs (OpenAI GPT-4 Vision + Anthropic Claude) in a single production pipeline, combining multimodal image analysis with structured report generation to produce complete adjuster-ready claim summaries.

EDUCATION

Master of Science in Computer Science | State University of New York at Buffalo, Buffalo, NY May 2025

Relevant Courses: Machine Learning, Deep Learning, Computer Vision and Image Processing, Data Intensive Computing,

Statistical Learning and Data Mining, Data Models and Query Language, Programming and Database Fundamentals for

Data Science, Operating Systems and Computer Security.